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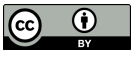
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Full paper

Telecenters for the Future in Tea Estates of Sri Lanka

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Abstract. This paper reports on a study conducted at one of the Sri Lankan tea estate districts, exploring the present day status of telecenters to examine how they have succeeded in meeting the initial high expectations attached to them. During a field study, two major types of telecenters have been examined through observations, interviews and document analysis. Our findings suggest that the challenges of the initiation phase still prevail. The hopes are placed on the younger generation, as they are regarded as those who can benefit from the ICTs and thus contribute to the development of the remote communities of tea estates. In the concluding discussion, we advocate for the possibilities of co-designing new services that might help to transform the telecenters to meet the needs and requirements of the tea estate communities of today and tomorrow.

Keywords. Digital Divide · Education · Telecenters · Tea estate areas · Accessibility · Sustainability · ICT literacy · Community development · Civic services.

1 Introduction

This paper reports on a study conducted in one of the Sri Lankan tea estate areas, Nuwara Eliya, and discusses the present-day telecenter situation. During the field study, two major types of telecenter arrangements were identified: the government initiated Nenasala telecenters, and the e-kiosk type community centers run by the Thondaman Foundation connected to the Ceylon Workers' Congress. The purpose of this paper is to describe and analyse the major telecenter models that exist today, and discuss the potential and challenges of these to provide e-services to their communities.

1.1 About the Telecenter Movement

The telecenter movement has its origins in Scandinavia and the United States, where telecenters were established in the 1980s in an effort to provide access to Information and Communication Technology (ICT), particularly in rural areas where the access to the new technology was limited [1].

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Telecenters have many names and many different forms depending on where and by whom they have been established. In general terms, telecenters can be defined as physical places which facilitate public access to ICTs, especially to the Internet, and thus benefit educational, personal, social, and economic development [2]. Telecenters have varying models for administration and funding. They can be run and financed by NGOs, government, international organizations, development banks, foreign assistance agencies, private companies and individual entrepreneurs [3]. Despite the various names – telecenters, e-kiosks, and Internet cafés – the common denominator is the aim to bring ICTs to the inhabitants of a certain area. Another name for telecenters is community resource center and they have been established in many countries with the purpose of assisting to decrease the digital divide [4, 5]. In Sri Lanka, the Information and Communication Technology Agency (ICTA) introduced telecenters in 2004. Over the years, some 838 Nenasala telecenters have been established around the island in rural and semi-rural regions [6].

1.2 Context of the Study: Tea Estate Areas in Sri Lanka

Estate areas refer to the coffee and later tea and rubber plantations originally established by the British in the nineteenth century. Geographically they are concentrated in South-central Sri Lanka. Tea is an important part of the Sri Lankan economy and the estate areas developed their own particular socio-economic and cultural conditions with the migrated Indian Tamil population as the workforce. The plantation workers reside within the boundaries of the tea estate, earning their livelihood in the tea factories or on the steep slopes of the tea plantation plucking tealeaves. Estate areas have remained isolated and marginalized with limited possibilities for social mobility and development.

According to the year 2012 census, there are about 900,000 people living in the tea estate areas of Sri Lanka [Calculated based on 7], which is 4.4% of the total population. The majority of these, 84%, live in the following six districts: Nuwara Eliya, Badulla, Rathnapura, Kegalle, Kandy and Matale. Nuwara Eliya is the largest district with 53% of the total tea estate population. Access to education and societal services is limited [8] within the tea estates. Even though the literacy rates for the three population groups – urban, rural and estate populations – are almost even, there is a clear difference concerning computer literacy. The estate population reaches only to 8.8%, which is considerably lower than the 36.8%, the corresponding figure for the urban population [8]. In this study, computer literacy denotes basic computer skills, while digital literacy refers to more in-depth skills and competences in ICT use and awareness of potential opportunities with ICT services at the telecenters.

Despite the government efforts, the internal digital divide is still remarkable in Sri Lanka separating the urban areas where access to ICTs and the Internet is relatively good, from the rural areas and particularly tea estate areas, where the access to ICTs is poor [8]. Accessibility issues are related to both technical matters such as weak infrastructure, low bandwidth and lack of personal computers or smartphones, and also to lack of basic computer skills. Other important aspects are the lack of relevant content in vernacular languages as well as issues related to special needs for people with im-

pairments [9]. Despite the investment of the Sri Lankan government in building infrastructure through telecenters [1], the telecenter network does not always reach those who would need it most: the underprivileged population of the tea estate areas.

By establishing telecenters, a particular kind of ICT infrastructure is being brought into a community. This infrastructure may have potential of empowering the members of the community through reducing isolation, bridging the digital divide, creating educational and economic opportunities [10] [11] [12]. In official (government) rhetoric [13] this potential is often presented as a matter-of-fact solution to the various issues facing isolated rural communities. This is the case also with the nationwide Nenasala project, introduced for strengthening democratic processes, enhancing socio-economic development and quality of life for those who are economically and socially marginalized in the society [1]. How this transformation and empowerment is actually taking place, if it does so, needs to be examined, as our understandings of the dynamics between communities and new technologies still are limited.

Now that more than ten years have passed since the founding of the first telecenters, it is time to revisit the Sri Lankan telecenter landscape to see how the centers are functioning today and, most importantly, how they have managed to meet the high expectations of transformation and empowerment of the rural population that were attached to them in the beginning. It is also of interest to explore the present day telecenter landscape as new actors have appeared. The dynamics between the communities and ICTs needs to be analyzed as this can inform us of the ways of improving the technologies for the good of the people.

1.3 Research Problem and Question

Telecenters were introduced in Sri Lanka with the hope of providing access to ICTs for the population and thus contributing to the improvement of people's lives. The problem is that the potential of telecenters to empower members of communities is often presented as a matter-of-fact in government rhetoric and independent studies of the impact of the centers are insufficient. This is why we wish to examine the present situation focusing on the following questions: What kind of telecenters are there today in tea estate areas of Sri Lanka and how are they used by the local population? The answer to this question is important as a baseline for further research on the relation between tea estate population and the ICTs and how the services could be developed.

The rest of the paper is structured as follows: Next section describes the research methodology applied in the study, followed by our findings in the form of an analysis of the tea estates and of the two main types of telecenters, which we found during our field visits. We also account for the attitudes and the actual use of present-day telecenters. In our discussion, we highlight the issue of the future role of telecenters and present the specific opportunities and challenges that we have identified in our study. We conclude by presenting our ideas for further research in the form of participatory co-design of future telecenters.

2 Method

In this study we have applied exploratory case study strategy [14] where the purpose is to learn about a particular, relatively unknown phenomenon – in this case – Sri Lankan telecenters in their present day form in the specific geographical area of tea estates. We chose the Nuwara Eliya district because it has got the largest population among the tea estates. As we wanted to examine what kind of telecenters there are today and how they are used by the local population, we chose to combine various data collection methods. We carried out content analysis of existing ICT policy documents and government and private actors' websites related to telecenters to get an official and general picture of the situation, and the types of telecenters that are in function. We also visited the Nuwara Eliya district to learn about the actual conditions of telecenters. We interviewed telecenter managers and instructors. We conducted group discussions with the users and made observations at a number of telecenters. In total, the authors visited four tea estates and seven telecenters in these tea estates. We interviewed 10 telecenter instructors, the district coordinator of the Thondaman Foundation telecenters, four tea estate managers and groups of telecenter users. The seven telecenters were those who responded to our request to visit and interview them. This random sample covers the government initiated Nenasala telecenters and the e-kiosk type community centers run by Thondaman Foundation.

3 Findings: Present-day Tea Estate Telecenters

At present, there can be found two major types of telecenters: those initiated by the Sri Lankan government called Nenasala and those initiated by the Thondaman Foundation called Prajashakthi. In this section, we give a short account of both types of telecenters that we visited during our field study.

3.1 Nenasala Telecenters

Nenasala telecenters were introduced in 2004 by the Sri Lankan government as part of the eSriLanka project. The original purpose of the telecenters was threefold: 1) to contribute to the so-called diffusion effect which is about developing people's use of ICTs; 2) to bring in the technology companies for the provision of connectivity and technology; and 3) to stimulate other sectors in society to use ICTs for the development of their services to the communities (for example healthcare, education, agriculture, and retail) [1].

The government provided telecenters with a 'start package' of basic equipment and partial financial support during the first four years after which the telecenters should be self-sustained. The government would also subsidize connectivity over a period of four years [13] [1]. After this time the center would take over.

The goal of the Nenasala project was to establish multi-service community information centers. The centers were also envisioned to function as a hub of local, national and global information resources and thus be part of the efforts in poverty allevia-

tion, social and economic development and peace building. Furthermore, the goal was to make these centers economically and technically sustainable in order to ensure long-term development.

Right from the beginning, there existed different kinds of models concerning ownership, location, and economic arrangements. Four types of Nenasala telecenters can be distinguished; telecenters were established in conjunction to: Buddhist or Hindu temples (temple model); libraries (library model); non-government organizations (NGO model); or as a business in a form of Internet cafes (enterprise model). According to personal communication with an ICTA representative on May 29, 2015, the enterprise model never gained much popularity, as only one person would make profit on such an activity. The library model is gaining more popularity at present. Also, the ICTA representatives believe that this is the most viable model because libraries are already established services in a community and they provide a location and facilities where to develop ICT related activities and services. Libraries are run by the local councils and thus provide a sustainable solution even for telecenter activities. However, while the library model seems more sustainable, they are new and just starting. Only through time we can say how they are succeeding.

Nenasala telecenters have been a major investment for the Sri Lankan government with striving goals for the program. From year 2004 and onwards, the ambitious plan has been to install 1000 telecenters around the country (at present 838) [6]. However, it is difficult to find information about the number of telecenters that are actually in function today. In our efforts to contact telecenters, we found that several of them were closed down.

3.2 Prajashakthi E-kiosks

Savumiamoorthy Thondaman formed the Ceylon Workers' Congress (CWC) as an Indian Estates Workers Trade Union in 1950. It became the largest trade union in the country. It has traditionally represented Tamil population (Sri Lankan Tamils of Indian origin) in the plantation sector of the economy and has a strong position among the tea estate workers. Thondaman has been the dominant trade union leader of the Indian Tamil workers for nearly four decades. The work of the Savumiamoorthy Thondaman Memorial Foundation is closely connected to the Millennium Development Goals fighting poverty and providing education and health care. The Foundation took the initiative to e-kiosk centers with the overall vision of achieving socio economic development through ICT and involvement of all stakeholders [15].

There are high hopes and expectations attached to ICTs. Computer literacy is seen as the key competence for the plantation youth in becoming active citizens in a new knowledge society. In his address on the occasion of the opening of an e-kiosk center, the former president Rajapaksa expressed his hopes of how the center would assist empowering the plantation community by providing access to knowledge through the use of ICT [16].

The e-kiosk title "Prajashakthi" translates to 'community empowerment', which is the central idea of the e-kiosk center project. The marginalized tea estate communities are to be empowered with the help of Internet and ICTs. With its political roots, the

Foundation seeks to transform the communities into well-informed, self-sufficient parts of the Sri Lankan society.

The Prajashakthi telecenters aim at operating as a holistic model of service provider – a community center that brings different groups to the center – especially children, youth, women, people with impairments – for various activities that serve the purpose of empowerment. Computer literacy is part of the services but it is combined with other aspects like English language learning, entrepreneurship and health information. Telecenters address the social and cultural needs of the society encouraging children and youth to express themselves through writing, song and dance thus reproducing the local culture but also finding novel ways for cultural utterances.

Microsoft Office and Internet browsing are central learning tasks. Through the community website www.thondaman.org, the children are given email and chat facilities. Sports and physical education also form an important part of activities in a community center of Prajashakthi type. The Foundation organizes common training seminars for the telecenter instructors and develops own educational content, which is shared by the centers. All the services offered by these centers are free of charge, which is not the case in Nenasala telecenters where the fees make a basic income for the managers and need to cover the expenses of the telecenter.

In summary, the two types of telecenters described here differ in several aspects. Nenasala telecenters are a state initiative whereas Prajashakthi telecenters have been founded through a private actor. The Nenasala telecenters were given a start package but are to be self-reliant subsequently. Prajashakthi telecenters are financially supported by both the Thondaman Foundation and the government. The approach of the Foundation is a wider social, cultural and economic engagement than that of the Nenasala telecenters, which only focus on ICT accessibility and training.

3.3 Attitudes Towards and Use of Telecenters

The tea estate managers describe the relation to the estate workers in a feudal manner. As one manager said: "We take care of them from the cradle to the grave". Housing, medical services, schooling, shops are organized by the estate. The estate managers are well aware of the existing co-dependency between the estate population and the tea industry: if the tea estates wish to keep their employees, they need to provide good working and living conditions. The development of the tea industry goes hand in hand with the development of the workers.

The estate managers, which we interviewed, welcome ICT and education; they consider well-educated workers as good workers with potential to keep up with the new requirements related to the (technological) development of the tea industry. Some of the tea estates have even established estate owned telecenters where workers and their families are able to learn about the use of ICTs and also get language training in English language.

Although telecenters are thought of as open for the whole population, it is the children, youth and the younger generation that mainly use them. Gamage and Halpin [17] report that there is a general belief of telecenters being only for educated people. This belief has limited the use of the centers. In a sense, this is also what we found

during our field study. ICTs are considered as something for the younger generation. In our discussions with some families, the parents told us that it is important for the young to learn about ICTs so that they can get hold of new opportunities and advance in their lives.

As to the parents themselves, telecenters are not considered as places of services or opportunities. This is also the attitude of the telecenter instructors: they focus on the younger generation and have designed their activities to meet the needs of the young by providing for example special software for drawing and educational games such as Typing Master on telecenter computers. We visited one Nenasala telecenter where they planned to start ICT lessons for (young) mothers, who would then spread the skills further to their children and other family members.

The community centers of Thondaman Foundation with their holistic approach to community development address even other groups with their activities. For example, they provide sewing classes for women and they support local entrepreneurship.

Instructors and users that we spoke with expressed their wish to learn even other skills – for example image processing. However, the software e.g. Adobe Photoshop is costly and cannot be purchased by the telecenters. The knowledge of open source alternatives seems to be limited among telecenter managers, which also hinders the development of services.

As English language is a second language for the population, language training is considered important and thus provided in telecenters. There is, however, lack of suitable, contextualized training programs for these Tamil speaking populations of tea estates. English language is even a key to a lot of information on the Internet and, consequently, poor English language skills form a barrier for the tea estate population to use the ICTs and the Internet [8].

Telecenters are also a place where the younger generation get together and socialize even in a more leisurely manner. Smaller children make drawings with the help of computers often several of them sitting together around a computer looking at and commenting each other's work. Young people play entertainment games such as Angry Birds with Android-based smartphones in case they have one and some also use Facebook and Twitter.

At present, no governmental e-services are provided through tea estate telecenters even though e-Sri Lanka project has been successfully implemented in many other parts of the island. It seems that both the telecenter managers and instructors are not aware of the potential opportunities of various e-services that might improve the local living conditions. For example, health care information about disease prevention could be very useful for the tea estate community. However, there does not seem to exist any kind of framework for such e-service. The telecenter staff also lacks the resources for further improvement as they are struggling to keep the basic services going. This was noted at telecenter manager interviews in May 2015.

When asked about the present use and the services provided, telecenter managers and instructors point out the limited resources and competences as an obstacle. If they had better, functioning equipment, and a better variety of software and knowledge of the software, they would be more attractive to the users. At a Nenasala library telecenter with about 100 active members, the manager said: "Our users would like to

learn web design, but we don't have a good knowledge". In addition, there seems to be lack of collaboration between different actors in the educational sector that would help to consolidate the services and bring more users to the telecenters. As one of the Prajashakthi e-kiosk managers suggested: "We should have a (national) system to give a certificate for our participants in collaboration with a university or another institution". A certificate would give a needed recognition of the (training) courses taken at the telecenters, and it could be used when applying for jobs. Collaboration with well-recognized educational institutions in society, both for providing content as well as for validation of knowledge, would be important for the telecenters. The e-kiosk manager continued: "If we can get theory from a university, we can start teaching. Thondaman vocational training is registered, these centers are not registered", suggesting that with a proper license, telecenters could become a competitive and attractive actor in the educational sector of the country.

Our findings suggest that the attitudes towards and the use of telecenters are constrained by the limited ICT resources and knowledge as well as limited awareness of the possibilities.

4 Discussion

Even though the idea of telecenters is being questioned as mobile phones are becoming the possession of everyone [18], research also shows that telecenters offer important complementary services providing access to technology and information, as well as playing a role in the development of ICT skills [19]. In Sri Lanka, the government envisions free Internet via Wi-Fi for the next generation of Nenasalas, which we learnt at a meeting with ICTA in May 2015. This would certainly enhance the possibilities of providing new services through telecenteres.

Future telecenters can play a central role as 1) a distribution node for education, 2) providing peripherals and 3) technical support services: 1) Basic computer training and use of simple services such as e-mail, is necessary for the 91.2% of the estate sector [8] that still lack computer literacy to enable use of ICT services such as an "English for all" application [20]. For those who are digital literates, courses in more advanced skills could be provided in collaboration with the Sri Lankan universities, e.g. with Wikiversity [21]. These services could also be transferred to smartphones or USB sticks at the telecenters, to bring home where Internet connection may not work; 2) While smartphones provide many opportunities, the small format of the screen and touch based interface make them hard to use for all tasks. To write a longer document, to program a computer or to make more advanced graphics require peripherals. Instead of (or complimentary to) personal computers, telecenters could provide larger screens, printers, and keyboards etc. that visitors could connect to with their smartphones; and 3) When a computer or mobile phone breaks down either in software or hardware, it is necessary to have someone nearby who can help to fix the problem.

The Prajashakthi e-kiosks seem to benefit from the holistic approach to social reform and transformation that is part of the foundation of the CWC. The e-kiosks have

several different activities of which ICT related ones are only a part. The instructors have continuous training and competitive salaries, which increases their motivation and dedication in their work in e-kiosks. The Nenasala telecenters have not included the possibilities of entrepreneurial activities for the estate inhabitants as part of their mission. The Prajashakthi e-kiosks recognize the importance of economic self-reliance and advice the inhabitants within the tea estates in the process of establishing small-scale businesses.

The future of telecenters is closely dependent on the attitudes of the users and the telecenter staff. Awareness building, information and communication about the novel technical possibilities is crucial [22] [23]. For this reason, it would be very important for the (Nenasala) telecenters to network and exchange information, experiences, knowledge and resources between the managers and instructors. Continuous, active interaction with the local community is also vital for raising awareness. Here, the use of vernacular languages plays a part. Recognition of the local language (Tamil) is important for creating participation, ownership and empowerment among the community members. Increased awareness can subsequently lead to creation of relevant, contextualized educational content and creation of new services.

ICT developers and designers can play an important role in informing of the novel technical possibilities and contribute to the development of new services in a respectful collaboration with the community. The appropriation of ICTs can thus take place as self-organized activities of the community rather than an outside intervention. It is the community that must define the needs and requirements that the future services are based on.

In addition to the awareness and competence issues, telecenters seem to struggle with the same questions as reported before [17] [24]. Adequate infrastructure, stable Internet connectivity and power supply are the very base of any telecenter operations. If this base does not exist, it will be very hard to develop and provide new services. Sri Lankan government is now launching the Google Loon [13], which is to provide Internet connection to the island. Finding a suitable economic model for the telecenters [3] is also an issue that still is searching for a sustainable answer.

5 Conclusions and Future Research

The purpose of the study presented in this paper has been to examine the present situation of telecenters in the Sri Lankan tea estate areas; what types of telecenters exist and how the local population uses these. We have found two major actors in the telecenter field: the government initiated Nenasala telecenters and the Prajashakthi telecenters founded by the Thondaman Foundation. The conditions for these two types of telecenters differ in several respects: funding arrangements, management (salary, training, competence development), range of services, equipment, and infrastructure. These conditions frame their possibilities to carry out their mission purposefully, as well as their potential of being part of the future transformational processes. Our previous studies show that communication-intermediary tasks, networking between telecenters and stakeholders, and local knowledge and practices are important for gaining

sustainable transformations. These are issues that telecenters need to address for their further development.

We believe that telecenters could continue to play a relevant role despite the widespread use of smartphones that have their limitations, too. A center that can support the whole community with stable Internet connection, relevant educational programs not only in ICT use but also in other areas as well in close collaboration with other educational institutions, and that provides necessary technological support, would seem like a viable future model. However, to move beyond basic skills training of today, both the instructors and the users must become aware of potential ICT opportunities and services. Based upon this study, we believe that raising such awareness could be achieved through application of participatory action research design. In such a process, new services may be co-designed based upon the needs and requirements of the local population. These new services can play an important role in transformative processes leading to improvement of lives in the remote, disadvantaged communities of tea estates.

References

1. Shadrach, B.: Nenasala The Sri Lankan Telecentre Experience. Information and Communication technology Agency, Colombo (2012).
2. Reilly, K., Gómez, R.: Comparing Approaches : Telecentre Evaluation Experiences in Asia and Latin America1. Electron. J. Inf. Syst. Dev. Ctries. 4, 1–17 (2001).
3. Windsor, S.S., Royal, C.: Different telecentre models in ICT for development and their impact on organizational sustainability. Int. J. Technol. Manag. Sustain. Dev. 13, 161–175 (2014).
4. Jensen, M., Esterhuysen, A.: The Community Telecentre Cookbook for Africa Recipes for Self-Sustainability. How to Establish a Multi-purpose Community Telecentre in Africa. United Nations Educational Scientific and Cultural Organization, Paris (2001).
5. Islam, M.A., Tsuji, K.: Bridging digital divide in Bangladesh: study on community information centers, (2011).
6. Nenasala Official Web Site: Nenasala, http://www.nenasala.lk/.
7. Department of Census and Statistics Sri Lanka: Department of Census and Statistics, Sri Lanka, http://www.statistics.gov.lk/PopHouSat/CPH2011/index.php?fileName=Activities/TentativelistofPublications.
8. Peiris, R., Mozelius, P., Männikkö-Barbutiu, S., Westin, T.: Bridging the Digital Divide in Sri Lankan Tea Estate Areas. In: IFIP 9.4 13th International Conference on Social Implications of Computers in Developing Countries. , Negombo (2015).
9. Gomez, R.: Libraries, telecentres, cybercafes and public access to ICT: International comparisons. (2011).
10. Mishra, G.: Telecentres as a Medium for Good Governance in Rural India. In: Handbook of Research on Democratic Strategies and Citizen-Centered E-Government

Services. pp. 56–72. IGI Global (2014).
11. Malek, J.A., Ahmad, A.R., Awang, M.M., Alfitri: Symbiotic relationship between telecentre and lifelong learning for rural community development: A Malaysian experience. Turkish Online J. Educ. Technol. 13, 148–155 (2014).
12. Okello-Obura, C., Kacunguzi, D., Kidaaga, J.: Application of information and communications technologies in business operations and access to business information by small and medium enterprises in western Uganda, http://reference.sabinet.co.za/webx/access/electronic_journals/mousaion/mousaion_v33_n1_a8.pdf, (2015).
13. ICTA: Information and Communication Technology Agency || Information and Communication Technology Agency of Sri Lanka - ICTA, http://www.icta.lk/index.php?lang=en.
14. Yin, R.K.: Case Study Research: Design and Methods. SAGE Publications, Thousand Oak CA (2003).
15. About Thondaman Foundation: Thondaman Foundation, http://www.thondaman.org/aboutstmf.html.
16. Saumyamoorthi Thondaman Memorial Foundation: Prajashakthi. Saumyamoorthi Thondaman Memorial Foundation, Colombo (2007).
17. Gamage, P., Halpin, E.F.: E-Sri Lanka: bridging the digital divide, (2007).
18. Central Bank of Sri Lanka: Annual Report 2014. , Colombo, Sri Lanka (2014).
19. Sey, A., Coward, C., Bar, F., Sciadas, G., Koepke, L., Alampay, E., Best, M., Blake, T., Donner, J., Gordon, A.: Connecting people for development : why public access ICTs matter. 241 (2013).
20. Google play: English for All, https://play.google.com/store/apps/details?id=com.efa.learner.intermediate.
21. Wikiversity:FAQ - Wikiversity, http://en.wikiversity.org/wiki/Wikiversity:FAQ#What_is_Wikiversity.3F.
22. Sulaiman V, R., Hall, A., Kalaivani, N.J., Dorai, K., Reddy, T.S.V.: Necessary, But Not Sufficient: Critiquing the Role of Information and Communication Technology in Putting Knowledge into Use, (2012).
23. Islam, M.S., Hasan, M.N.: Multipurpose community telecentres in Bangladesh: problems and prospects, (2009).
24. Jensen, M.: Nenasala Review on behalf of the World Bank Final Report. ICTA. (2007).